PadCare Labs

Complaint Management in PadCareOne

Product Analyst (Process & Systems) — take-home assessment

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The design in five lines

- **Every issue becomes a Case** logged by whoever receives it, owned by the account's CS owner from the first minute.
- **Four case types** (operational, billing, service request, account-level) chosen by five ordered questions; a **status check is attached to its original case**, never a new complaint.
- **Severity follows impact** (hygiene exposure now, service or billing commitment broken, repeat, scope), never who is asking.
- **Ownership moves only when the receiver accepts**; the customer always hears from one CS owner.
- **"Closed" needs evidence and the customer's confirmation**; returns within 14 days reopen the case.

How to read this document

- `[A-xx]` = assumption (Section 2). All SLA targets and thresholds are **proposals** to validate with PadCare.
- ⚠️ = a problem found while walking a complaint through the design (Task 2). † = Task 1 rule changed because of it.
- Supporting working files (full field list, 34 validation rules, data model, metric definitions, audit) are available on request.

1. Problem and scope

PadCare cannot reliably answer four questions about any customer issue: **did we receive it, who owns it right now, is it actually fixed, has it happened before?** — nor, across issues, **where are we failing most?** The eight challenges reduce to four root causes, and the design targets those:

Root cause	Challenges	Design response
Issue lives with a person, not a system	1, 4, 7	Central register; receiver logs; owner from creation; tracked handoffs
Every interaction is called a "complaint"	3, 5, 6	Case types with their own owner, path and evidence of "done"
No memory of history	2, 6	History panel at intake; structured customer / site / category
No shared meaning of "serious" or "done"	5, 8	Impact-based severity; customer-confirmed closure

In scope — v1	Deferred (why)
Central case register inside PadCareOne; manual logging from all 6 channels Classification, related-record search, impact severity, routing, handoffs, linked tasks Lifecycle with evidence, customer confirmation, reopen CS-sent customer updates with a "maximum silence" timer Structured fields and an event log for reporting	WhatsApp / email integration — none evidenced `[A-04]`; prove the process first Customer self-service portal — PadCareOne is internal `[A-06]` Invoice issuing, field dispatch — existing processes; the module links to them Automatic (AI) classification — needs the labelled history v1 creates

2. Assumptions

Facts used come only from the brief. Everything below is assumed; the right-hand column says what changes if it is wrong.

ID	Assumption	Validate with	If wrong
A-01	PadCareOne holds customer and site master records	Product Lead	Validation and repeat detection fall back to free text
A-02	PadCareOne holds the service schedule per site (date, status)	Product / Ops	Missed-collection cases cannot link to a visit
A-03	Invoices are not in PadCareOne; CS enters the invoice number, Finance verifies	Finance	v2 could look invoices up
A-04	No channel integrations; the receiver logs manually	Product Lead	Integration reduces effort, flow unchanged
A-05	SEs have a mobile view to log issues and record visit evidence	Ops	SE calls CS, CS logs
A-06	No customer portal; CS updates customers on their usual channel	Product Lead	Portal replaces the silence timer
A-07	No contractual complaint SLAs exist; all targets are proposals	CS / KAMs	Contract terms override
A-09	Every account has a named CS owner; a city CS queue covers leave	CS	Routing needs a queue owner
A-10	Only large accounts have a KAM; otherwise the Ops City Manager handles account concerns	Sales / KAM	S-05 routing changes
A-11	Only Finance issues corrections and credit notes; Finance has an escalation contact	Finance	—
A-12	Business hours 09:00–18:00 Mon–Sat; Hygiene-Critical runs on calendar hours	Ops / CS	SLA arithmetic shifts
A-16	An additional unit may change charges, so it needs commercial approval	Sales / Finance	Approval step dropped

ID	Assumption	Validate with	If wrong
A-18	Owner of site master data unknown; default = Ops City Manager of the city	Product Lead	S-02 correction task re-routed
A-20	An SE can reach a site within 24 h in most cities	Ops City Managers	Hygiene-Critical target varies by city

3. Challenges Map

Step numbers refer to the flow in Section 4. ● addressed in v1 · ◐ partly (limitation stated).

#	Challenge	Design choice that addresses it	Where	
1	Complaints lost between channels	Whoever receives it logs it (≈2 min). Owned by the account CS owner from creation; city queue covers leave. A status check about an unlogged issue creates it, flagged *not logged at receipt*	1.1–1.2, 2.2	◐
2	Repeat vs new unknown	History panel before classification. Repeats suggested from the 2nd case; the 3rd in 90 days is flagged Recurrence: severity up, root-cause note required	2.1, 2.4	●
3	Everything called a "complaint"	4 case types chosen by 5 ordered questions; status check = follow-up on its parent; mixed messages split into linked cases	2.2–2.3	●
4	No clear handoff	Handoff with reason + checklist; sender accountable until the receiver accepts or returns; silence escalates; other teams' work as linked tasks	3.1–3.4, 4.3	●
5	"Resolved" ≠ resolved	Evidence required by type before Awaiting Confirmation; closure needs the customer (or a defined no-response rule); return ≤14 days reopens	4.4, 5.1–5.4	●
6	No patterns visible	Structured city / site / type / sub-category / severity + an event for every change → e.g. category share by city	ST6, §7	●
7	No customer visibility	CS updates at fixed points + "maximum silence" timer; chasing is counted	2.6, R3–R6	◐
8	Severity by loudest voice	System proposes severity from impact answers; seniority, tone, revenue are not inputs; overrides need a reason and are reported	2.5	●

Limitations (◐): 1 — still depends on people logging; lost issues become *visible and counted* when customers chase, and v2 inbox/WhatsApp capture removes the dependency. 7 — no self-service in v1; visibility depends on CS discipline, which the silence timer measures; v2 adds a read-only status link.

4. Task 1 — End-to-end complaint flow

4.1 Classification: what kinds of "complaint" exist

The five samples are traps if read by keyword: #4 mentions a missed service, #5 service quality, #2 a site move, #3 overflow. Classification therefore asks **what the customer needs PadCare to do**, in a fixed order — the first "yes" wins:

Order	Question	Result	Trap it prevents
1	Asking about an issue already raised?	Status check → follow-up on parent	#4 becoming a duplicate complaint
2	Wants an account-level response, or reports a trend across sites?	Account-level concern	#5 becoming a vague complaint with nothing to fix
3	About an invoice, charge or credit?	Billing complaint	#2 logged as an address change Finance never sees
4	Fix something PadCare already committed to?	Operational complaint	—
5	Something new or changed?	Service request	#3 treated as a failure ("a service request in disguise")
+	Second independent need with a different owner?	Create a linked case	#3: overflow relief ≠ new unit

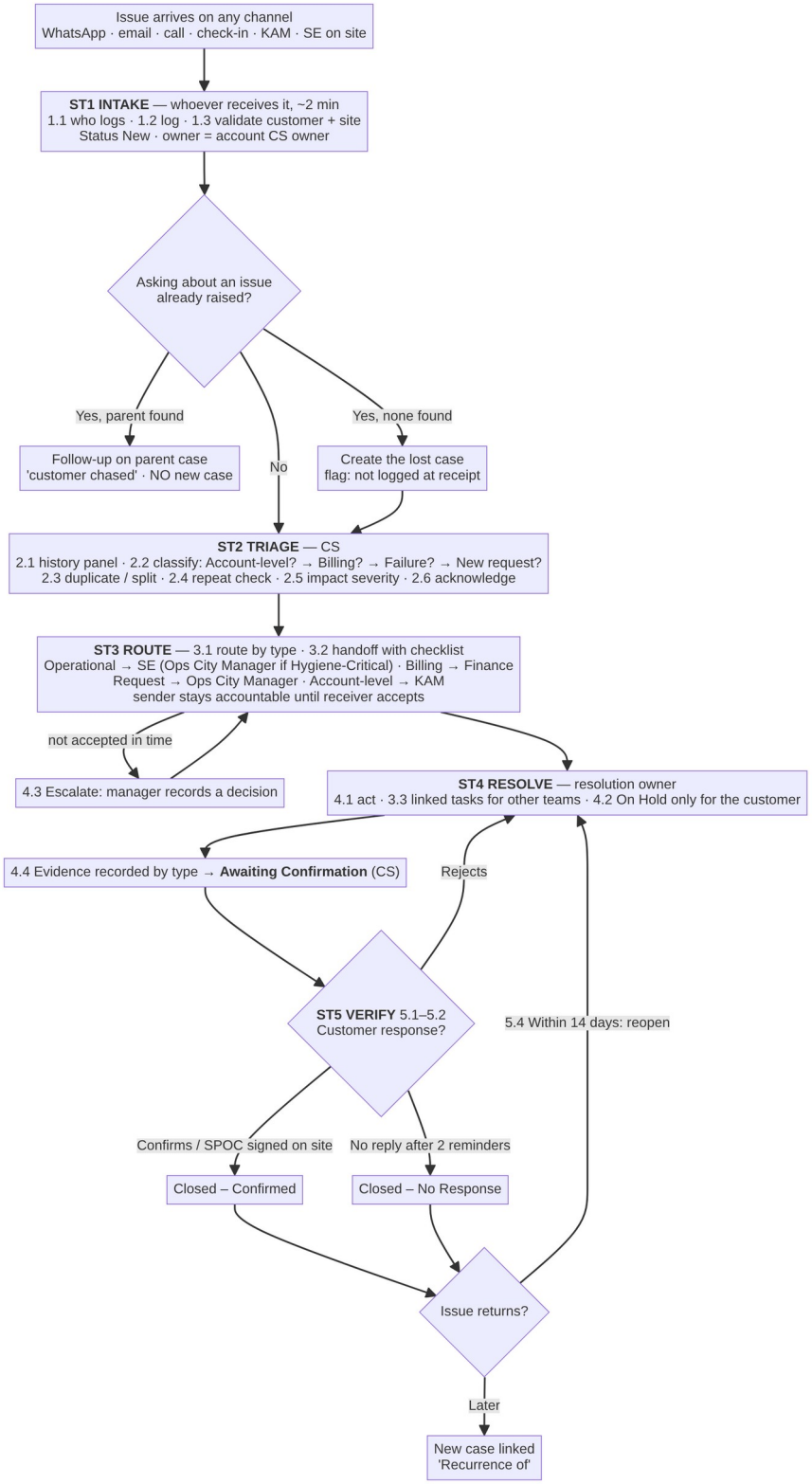
Type	Resolution owner	"Resolved" means (evidence enforced)	Samples
Operational complaint	Scheduled SE; Ops City Manager if Hygiene-Critical † (SE visit as a task)	Visit recorded + photo or SPOC sign-off; missed collection also needs the next visit back on the schedule †	#1, #3 (overflow)
Billing complaint	Finance (CS before and after)	All affected invoices reissued / credit note / documented explanation + any site-record task done †	#2
Service request	Ops City Manager (+ KAM approval task if charges change)	Fulfilled (e.g. installation recorded) or declined with a reason	#3 (new unit)
Account-level concern	KAM (else Ops City Manager)	Meeting held; action plan with owners and review date; child cases for specific site failures	#5
Status check (not a case)	Parent's communication owner	Update sent and logged — never closes the parent	#4

4.2 Severity and priority — impact, not volume

Level	Observable test
Hygiene-Critical	A hygiene or health exposure exists at the site now: units full or overflowing, collection overdue, damaged or leaking unit
Service-Impacting	An agreed service was not delivered properly, without current exposure; or an invoice the customer cannot act on (wrong amount, or wrong entity / site / address) †
Low-Impact	No service, hygiene or financial consequence (cosmetic; invoice typo that does not stop processing)

- **Escalators (operational only)** † raise one level: 3rd case in 90 days at the site · same issue at 3+ sites · collection ≥3 days overdue. Billing never exceeds Service-Impacting — a billing repeat escalates to Finance and the KAM instead †.
- **Who decides:** the system proposes from CS's impact answers and the schedule/history; CS confirms. Overrides need a reason; lowering Hygiene-Critical is visible to the Ops City Manager. **Reporter role is recorded for communication only.**
- **Priority** is the computed queue order, not a second label: severity → time left on the SLA → flags (customer chased, resolution rejected, recurrence) → age. An *Expedite* flag exists for KAMs / Ops City Managers but needs a reason and is reported.

4.3 The flow (step numbers used throughout this document)



Statuses: New → Assigned (handoff not yet accepted) → In Progress → On Hold – Customer (only when waiting on the customer; clock paused) → Awaiting Confirmation → Closed – *reason* (Confirmed · Confirmed on site · Plan Agreed · No Response · Cancelled – Duplicate / Invalid). Reopen is an event, not a status. Nobody — any role — can close without evidence.

4.4 Routing, ownership and handoffs

- **Two owners:** a *resolution owner* who moves with the work, and a *communication owner* (CS) who stays — Finance can investigate without becoming the customer's contact.
- **Handoff = reason + checklist + due time.** CS → Finance needs invoice no., month, disputed item, expected outcome, and for a wrong site the correct address + effective date. The **sender stays accountable until the**

receiver accepts; the receiver may return it with a reason (back In Progress with the sender †). Not accepted in time → the receiver's escalation contact must **record a decision** (reassign / take over / new date) — no silent auto-reassignment.

- **Linked tasks** † for work that does not change who decides (SE visit, site-record correction, commercial approval). Mandatory tasks block resolution.
- **Escalation triggers:** handoff not accepted · SLA breached · 3rd recurrence · second customer rejection.

4.5 Closure and verification

- Evidence recorded → **Awaiting Confirmation**, owner back to CS, who tells the customer what was done and asks "is this resolved?". A sign-off by the reporter or SPOC on site counts as confirmation †.
- **Rejects** → back to the last owner, SLA keeps running from receipt; 2nd rejection escalates. **No reply** after reminders on day 2 and 4 → *Closed – No Response* at day 5, reported separately from confirmed closures.
- **Returns within 14 days** → reopen. Later → new case linked *Recurrence of*.

4.6 SLAs (proposed)

Clocks start when **the customer told us** (late logging stays visible) and pause **only** while waiting on the customer (internal waits stay visible). Response targets tell the customer where things stand; resolution targets measure the fix.

Target	Hygiene-Critical	Service-Impacting	Low-Impact / request	Why
R1 Log (received → logged)	1 h	2 business h	2 business h	Details are lost after a few hours
R2 Triage (logged → classified)	1 h	4 business h	1 business day	Severity is unknown until triage
R3 Acknowledge with case ref	2 h	Same day	Next business day	Before the customer thinks to chase
R4 Accept a handoff	1 h	4 business h	1 business day	Nothing sits for days (C4)
R5 First ETA	4 h	1 business day	2 business days	Customers chase when they have no ETA
R6 Max silence while open	1 day	2 business days	3 business days	Replaces a portal in v1 (C7)
X1 Operational fix	24 h (calendar)	3 business days	5 business days	Waste builds at weekends too; validate SE capacity `[A-20]`
X2 Billing fix	—	5 business days	7 business days	Corrected before the next invoice cycle
X3 Service request	Committed date set at triage (default 10 business days for installation)			A promised date is more honest than a generic target
X4 Account-level	Meeting ≤2 business days	Meeting ≤5; plan ≤10 business days	—	Plan needs the account's case history

4.7 Key decisions and trade-offs

I chose...	over...	because...	accepting that...
"Case" with 4 types; status check attached to its parent	One "complaint" record with a type dropdown	Each type needs a different owner and meaning of "done"; status checks have no work of their own	Teams learn a new word
Ordered classification questions	A flat dropdown	The samples are keyword traps (#2, #3, #4, #5)	A few seconds slower

I chose...	over...	because...	accepting that...
Two-step intake (fast Log, full Triage)	A complete form at first contact	A form slower than a WhatsApp reply will not be used (C1)	Records are briefly incomplete; R2 exposes stalls
Ownership on acceptance	Ownership on assignment	A forwarded email is today's "handoff"	Senders carry slow receivers; escalation limits it
Manager must record a decision on breach	Auto-reassign	Auto-reassign moves the problem, not the decision	Depends on managers acting (measured)
Auto-close after no reply, reported separately	Keep open until the customer answers	Cases would stay open forever	Some unresolved issues close; reopen window covers it

[DRAFT — rewrite in your own words] The flow answers what PadCare can't today: did we get it, who owns it, is it fixed, has it happened before. The receiver logs it and a CS owner holds it from creation (C1). Ordered classification keeps status checks, requests and account concerns out of the complaint queue (C3). A history panel and recurrence rule surface repeats (C2, C6). Ownership moves only on acceptance (C4), closure needs evidence plus the customer's confirmation (C5), and severity follows hygiene impact and recurrence, not the reporter's seniority (C8). The weakest areas are C1 and C7: v1 still depends on people logging and updating.

5. Task 2 — Two complaints walked through the design

Times are **illustrative** (not PadCare data) and checked against the SLAs. Both walks were run on the first version of the flow; every ⚠️ led to a fix now shown in Task 1 (†). Friction was added deliberately — a field SE who does not answer, Finance returning a case — because the happy path proves little.

5.1 Complaint #1 — "Collection was not collected this month… it's a hygiene issue."

Intake (CS who received the WhatsApp): customer, site, reporter (SPOC, facility manager), channel WhatsApp, received Mon 10:05, verbatim text + screenshot, guess = operational.

Time	Step	Owner · status	What happens	SLA
Mon 10:40	1.2 Log	CS · New	Logged in ~2 min; customer and site valid	R1 35 min ✓
10:45	2.1 History	CS · New	One earlier missed collection 50 days ago (repeat #2, no flag yet). Schedule: this month's visit *Rescheduled* with no date	—
10:50	2.2 Classify	CS · New	Not a status check, not account-level, not billing → fix a committed service → Operational / Missed collection. No duplicate, no split	—
11:05	2.5 Severity	CS · New	CS asks the SPOC: bins full on two floors → exposure yes; 7 days overdue (escalator) → Hygiene-Critical. The word "hygiene" triggered the question; the facts answered it	R2 25 min ✓
11:15	2.6 Ack	CS · New	WhatsApp: case ref, "urgent, visit time by 2 pm"	R3 1 h 10 ✓
11:20	3.2 Handoff	CS accountable · Assigned	To SE-1 (scheduled SE) with location, access contact, severity	R4 due 12:20
12:20	4.3 Escalate	CS · Assigned	⚠️ SE-1 on route, never accepted. Escalated to the Ops City Manager, who reassigns to SE-2 at 12:45	Decision 25 min ✓
12:55 → 16:50	4.1 Act	SE-2 · In Progress	SE-2 accepts; CS sends ETA 16:30 (R5 ✓); collection done on all floors; photo; SPOC signs in the SE app	—
16:55	4.4 Evidence	CS · Awaiting Conf.	Visit + photo + sign-off. ⚠️ Nothing checks next month's visit is back on the schedule	X1 6 h 50 ✓

Time	Step	Owner · status	What happens	SLA
17:30 → Tue 09:40	5.1–5.3	CS · Closed	⚠️ CS asks "resolved?" although the SPOC already signed on site. SPOC: "Yes, thanks" → Closed – Confirmed	✓

Classification why: an omission against a commitment (not a request); Hygiene-Critical on facts — the same result if a junior employee had politely reported it. **Ownership:** CS → (SE-1 never accepted) → Ops City Manager decision → SE-2 → CS. **Duration:** fixed in 6 h 50 against 24 h; closed in ~23.5 h. **Tested alternatives:** schedule shows *Completed* → data-integrity flag + Ops note; 3rd miss in 90 days → Recurrence + root-cause review; SPOC says "only 2 of 4 floors" → rejection, back to SE-2, clock still running.

5.2 Complaint #2 — "Your invoice for September shows charges for the Hyderabad site, but we shifted… in July."

Intake (CS on support-inbox duty — not the account owner): customer, old Hyderabad site, reporter (SPOC, admin), channel email, received Wed 11:30, email text, guess = billing. Owner set automatically to the **account** CS owner (C1).

Time	Step	Owner · status	What happens	SLA
Wed 13:00	1.2–1.3 Log	Account CS · New	Site exists but customer says it moved → Unverified site warning (does not block)	R1 1 h 30 ✓
14:40	2.2 Classify	CS · New	CS calls SPOC: services continue at the new site, amount right, but their accounts team cannot process an invoice with the old address → Billing / Wrong site-address	—
14:50	2.5 Severity	CS · New	⚠️ Two of my rules fit ("billed incorrectly" vs "document error, amount unaffected"). CS picks Service-Impacting because the invoice is unusable	R2 ✓
15:00–15:10	2.6 / 3.2	CS accountable · Assigned	Ack email with case ref; handoff to Finance with invoice no., month, disputed item, expectation	R3 ✓
16:30	3.2 Return	CS	⚠️ Finance returns: new address and effective date missing. Case sits in Assigned and the status model will not allow On Hold from there; CS needs a workaround	—
16:45 → Thu 10:15	4.2 Hold	CS · On Hold – Customer	Customer sends new address, effective 15 July; clock paused 2.5 business h	—
Thu 11:30	3.2 Accept	Finance · In Progress	Re-handed with full data; Finance accepts; CS sends ETA "Monday"	R4 1 h ✓, R5 ✓
Thu 15:00	4.1 Act	Finance	⚠️ July and August invoices also wrong — the customer named only September. Root cause: site record never updated. ⚠️ Finance has no in-system way to get it fixed without giving up ownership → emails the Ops City Manager (today's behaviour)	—
Mon 12:00	4.4 Evidence	CS · Awaiting Conf.	Three reissued invoice numbers + site-record change reference	X2 ~3.8 bd ✓
Mon 14:00 → Wed	5.1–5.3	CS · Closed	Invoices emailed; reminder day 2; customer: "Received, all fine" → Closed – Confirmed	✓

Classification why: the need is a corrected invoice → billing; the site record is PadCare's internal root cause, not a separate customer request. **Ownership:** account CS → (Finance returned) → CS on hold → Finance → CS. **Duration:** fixed in ~3.8 business days against 5; closed in 6. **Tested alternatives:** charges for a vacated site → credit note + Ops must explain visits recorded at an address the customer had left; Finance finds the invoice correct → documented explanation, two rejections escalate to the KAM.

5.3 What the walkthroughs changed

⚠️ Found	Why it mattered	Change (version)
Two severity definitions fit #2	Same case, different severity depending on who triages — C8 returns	Billing severity = "can the customer act on the invoice?" (v1 †)
Return left the case stuck in Assigned	A spec built literally blocks a real path	Return → In Progress with the sender (v1 †)

⚠️ Found	Why it mattered	Change (version)
Finance could only email for the site-record fix	Re-creates C4's forwarded emails inside the system	Linked tasks: owner + due date, case owner stays accountable (v1 †)
SE in the field did not accept within 1 h	The most urgent cases depended on the least reachable person	Hygiene-Critical → Ops City Manager owns, SE visit is a task (v1 †)
SPOC confirmed twice	Irritates the customer, delays closure	Reporter / SPOC sign-off on site = confirmation (v1 †)
Next visit never re-scheduled	Case closes while the cause continues	Missed-collection evidence includes the next scheduled visit (v1 †)
Finance returned for missing data	Each return costs ~a day	Address + effective date mandatory before Finance handoff (Task 3)
Only the named invoice would be fixed	Guarantees a second complaint	Finance checklist: all invoices since the effective date (v1 †)
Nothing verifies the next invoice	Fix assumed, not verified	Automatic next-invoice check (v2 — needs billing data)

[DRAFT — rewrite in your own words] Both cases met every proposed SLA, but walking them exposed gaps the diagram hides. Finance had no way to get the site record fixed without giving up ownership, so I added linked tasks. Finance's return showed the form must ask for the new address and effective date, and that my status model blocked On Hold after a return. A field SE could not realistically accept within an hour, so Hygiene-Critical visits now go to the Ops City Manager. And I had rated the same invoice error under two severity definitions, so I tightened the billing rule.

6. Task 3 — Intake form specification

One form, two steps. **L** = required to save the Log (kept to ~2 minutes) · **T** = required to complete Triage · **H** = required before the named handoff · **C** = conditional · **S** = system. Hard **errors** block; **soft blocks** force an explicit choice; **warnings** inform. No warning ever stops an issue being logged.

6.1 Key fields

Field	Type · Source	Notes & validation
Case ID	Auto ID · system	S. `PC-YYMM-NNNNN`; read-only
Channel	Dropdown · user	L. 6 brief channels + Other. Default by role (SE → "SE on site", KAM → "KAM escalation")
Received at	Date-time · user (default now)	L. Starts every SLA clock. Not future, ≤ logged time (error). Beyond R1 → "late registration" warning; >30 days → reason
Customer already acknowledged at	Date-time · user	Optional. Between received and now; counts for R3 only if a case ref was given
Customer account	Lookup · user	L. Active customers. Not found → "Unmatched customer" + name as stated; Triage blocked until matched
Site	Lookup filtered by customer · user	L (optional for account-level). Must belong to the customer — hard error. Not in list / moved → tick *Site unverified + site as stated* (warning); Triage must pick the right site or raise a site-record task
City, CS owner, KAM	Derived · system	S. From site / account; drive routing and reporting
Reporter + contact	Lookup or new contact · user	L. Phone (10 digits) or email required
Reporter role	Dropdown · user	L. SPOC / End user / Accounts / Leadership / Other. For communication only — never an input to severity
Customer's words	Long text · user	L. Verbatim, ≥10 characters; unchanged after triage
Classification Q1–Q5 → Case type	Yes/No questions · user; type derived	L as a guess, T confirmed. Change after triage needs a reason

Field	Type · Source	Notes & validation
Sub-category	Dependent dropdown · user	T. List per type (e.g. Billing: wrong amount, wrong site/address, charged for service not delivered, duplicate, credit note pending)
Parent case	Lookup · user	Status check only, L: this customer's open + recently closed cases. Or "No existing case" → full form, *not logged at receipt* flag
Related cases / links	System panel; link lookup + type · user	Open cases at site; same site + sub-category in 90 days. Same customer only (error). Open match → soft block "Same incident?"
Second independent need?	Yes/No · user	T. Yes → linked case pre-filled (#3)
Location on site	Text · user	T for operational and service requests ("4th floor washroom")
Scheduled visit · days overdue	Lookup · user; overdue · system	T for missed / late service. Schedule says *Completed* → data-integrity warning (#1). Overdue ≥3 days → escalator
Invoice number · invoice month	Text, month · user	T and H (Finance). 4–30 chars; month not in future; >12 months → warning
Disputed amount · expected outcome	Currency, dropdown · user	Amount >0 required for wrong amount / duplicate / not delivered. Outcome T (reissue / credit note / explanation / refund)
Can the customer process this invoice?	Yes/No · user	T (billing). No → Service-Impacting
Correct address · effective date	Text, date · user	H (Finance) when sub-category = wrong site/address or site flagged "moved" (#2). Date ≤ today; after invoice month → warning "invoice may be correct"
Quantity · requested-by · committed date	Number, dates · user	T (service request). Dates ≥ today; committed date is the X3 target
Current problem reported?	Yes/No · user	T (service request). Yes → soft block: create linked operational complaint or give reason (#3 "overflowing daily")
Sites affected · customer asks for	Multi-select, dropdown · user	T (account-level, #5). ≥1 site or "portfolio-wide"; meeting / review / commercial change
Hygiene exposure now?	Yes / No / Unknown · user	T (operational, account-level). Unknown cannot complete triage — "confirm whether units are full, overflowing or damaged" (#1)
System / final severity	Derived; dropdown · user	T. Hidden for service requests; billing max Service-Impacting. Override → reason (error); lowering Hygiene-Critical notifies Ops City Manager
Recurrence count · priority	Computed · system	S. 3rd case in 90 days → Recurrence flag
Attachments	Upload · user	JPG, PNG, PDF, EML, MSG, DOCX, XLSX; ≤10 MB; ≤5. A bad file is rejected alone — the form still saves

6.2 Conditions and dependencies

Required	Operational	Billing	Service request	Account-level	Status check
Channel, received, customer, reporter, words	L	L	L	L	L
Site	L	L	L	Sites affected T	L
Sub-category · severity	T · T	T · T (≤ S-I)	T · hidden	T · T	— · inherits
Type-specific	Location, scheduled visit, hygiene exposure	Invoice no./month, outcome, can process; address + date (H)	Quantity, dates, current problem	Sites, asks for, hygiene exposure	Parent case (L)

- **By channel:** SE on site → site pre-filled from the current visit, photo prompt; KAM escalation → note "reporter role does not affect severity"; check-in call → "add another issue from this call" keeps reporter and channel.

- Changing the customer clears the site and links and re-runs the site and repeat checks. Changing type hides (does not delete) type-specific fields until saved.
- **Handoff checklists** block the move: Finance needs invoice data (+ address/date for a wrong site); SE / Ops needs location, access contact, severity.

6.3 Exception cases

Status check (#4)

- Parent open → follow-up, priority raised, CS "update customer" task due within R3; **no new case, parent status unchanged**.
- Parent awaiting confirmation → ask if resolved. Closed ≤14 days and not fixed → reopen. Closed earlier → new case linked *Recurrence of*.
- No parent → the issue was lost: full form, *not logged at receipt*, received = customer-stated date. Parent of another customer → error.

Repeat complaint

- Open case, same site + sub-category → soft block: *same incident (follow-up)* / *same incident, second reporter* (saved as Cancelled – Duplicate, linked, reporter added to parent) / *different incident*.
- Resolved earlier, back within 90 days → suggested *Recurrence of* link; 3rd → flag, escalator, root-cause note. **Never merged automatically.**

Wrong site or customer

- Site of another customer → hard error. Site missing or moved (#2) → warning at Log, resolved at Triage (correct site or site-record task). Customer missing → "Unmatched customer"; triage blocked until matched.
- Wrong pairing found after triage → CS corrects with a reason; SLA clocks unchanged; repeat checks re-run.

Also handled: multiple issues in one message (split rule) · invalid attachments · unauthorised status changes (role matrix; closure without evidence impossible for any role) · customer non-response (reminders, then No Response) · reopening after closure.

6.4 Test cases — Complaint #2 (billing)

Data: customer *Acme* (fictional), old site HYD-014 (Acme), HYD-022 (another customer), invoice for Sep, move effective 15 Jul; CS-B logs from the inbox, CS-A owns the account, FIN-1 in Finance.

ID	Test	Steps	Expected result
TC-01	Valid billing complaint (positive)	CS-B logs the email (received 11:30), HYD-014. CS-A triages: Q3 yes; wrong site/address; invoice no. + month; outcome reissue; can process = No; address + 15 Jul; complete	Case ID created; status New; owner = CS-A, not CS-B; clocks from 11:30; type Billing; severity Service-Impacting; no hygiene question; suggested owner Finance
TC-02	Missing invoice information (negative)	(1) Complete triage without invoice no. (2) Add it; hand off without effective date. (3) Enter 15 Oct. (4) Correct to 15 Jul	(1) Blocked, stays New. (2) Triage completes, handoff blocked listing missing items. (3) Warning "move after invoice month", not blocking. (4) Handoff succeeds; no handoff event before (4)
TC-03	Invalid customer–site (negative)	(1) Acme + HYD-022; save. (2) Tick "customer says site moved" + stated address; save. (3) Complete triage. (4) Pick HYD-014 + raise site-record task	(1) Hard error, nothing saved. (2) Saves with unverified warning. (3) Blocked until resolved. (4) Completes; mandatory task in Ops City Manager's queue
TC-04	Possible repeat (edge)	Open case -0031: same site, wrong site/address. Log the SPOC's email; choose "same incident — second reporter". Variant: 2 closed similar cases in 90 days	Soft block shows -0031; no auto-merge. New record Cancelled – Duplicate, linked; SPOC added to -0031; -0031 unchanged. Variant: Recurrence flag; severity stays Service-Impacting; Finance escalation + KAM informed
TC-05	Routing to Finance (positive + negative)	Hand off; FIN-1 returns ("need PO ref"); CS-A re-hands; no acceptance within 4 business h; FIN-1 accepts	Assigned with CS-A accountable; return → In Progress with CS-A; R4 breach escalates to Finance escalation contact; on accept owner FIN-1, communication owner still CS-A; never ownerless

ID	Test	Steps	Expected result
TC-06	Premature closure prevented (negative)	CS-A tries to close; FIN-1 tries Awaiting Confirmation without invoice ref, then with ref but task open; task done; customer asks "any update?"; FIN-1 tries to close; customer confirms	Each attempt blocked with the missing item; after task done → Awaiting Confirmation, owner CS-A; status check adds follow-up, status unchanged; FIN-1 cannot close; Closed – Confirmed only at the last step

[DRAFT — rewrite in your own words] The hardest decision was customer–site validation: block or warn? Blocking keeps repeat detection and reporting clean (C2, C6), but an issue that does not fit our master data would never be logged (C1) — and #2 is exactly that: the customer moved and our records did not. So a missing or disputed site is a warning at Log that must be resolved at Triage, while a site belonging to another customer stays a hard error, because that is a wrong pick, not missing data. The cost is records that are briefly unverified, which the short triage SLA limits.

7. Data that makes patterns visible (C6)

Five additions to PadCareOne (customer, site, contact and schedule are reused): **Case**, **Case link** (duplicate / recurrence / split / child), **Case task**, **Attachment**, and one append-only **Case event** log (every status, owner, handoff, customer update, follow-up, severity change and escalation, time-stamped). One log answers every metric and is the audit trail. Rules enforced in data: an open case always has both owners; a site must belong to the case's customer; a confirmed closure needs evidence; status checks never create cases.

Metric (all proposed — no baseline exists)	Definition	Challenge
Category share by city	Complaints of a sub-category in a city ÷ all complaints in that city (by received date) — answers "40% missed services in one city?"	6
Repeat rate	Complaints with ≥1 earlier same site + sub-category case in 90 days ÷ all complaints	2
Time in Assigned · return rate	Hours from handoff to accept/return; returns ÷ handoffs, by receiving team	4
Customer-confirmed closure · reopen rate	Confirmed ÷ valid closed; reopened within 14 days ÷ closed (cohort)	5
Chase rate · silence breaches	Cases with ≥1 status check ÷ cases; cases breaching R6	7
Severity overrides by reporter role	Overrides ÷ triaged, split by up/down and reporter role — is "loudest wins" creeping back?	8
Not logged at receipt	Cases created only when a customer chased ÷ all cases	1

Durations reported as median and 90th percentile; cancelled cases and status checks excluded from complaint counts.

8. Trade-offs, limitations and v2

Limitation in v1	Mitigation / next step
Logging still depends on people (C1)	Lost issues counted when customers chase · v2: capture from the support inbox, then WhatsApp Business
No customer self-service (C7)	Silence timer + chase metric · v2: read-only status link by case reference
Thresholds (24 h visit, 3 days overdue, 3 sites, 3rd in 90 days) are guesses	Calibrate after 2–3 months of module data; review overrides
More load on Ops City Managers (Hygiene-Critical ownership)	Validate volumes per city; a city dispatcher role if needed
Auto-closed "no response" cases may hide unresolved issues	Reported separately; 14-day reopen; SE check on next visit for Hygiene-Critical
#3 and #5 checked by design review only, not by a timed walkthrough	Next walkthroughs to run; #5 is the most likely to expose gaps (multi-city ownership)
Next invoice after a billing fix is not verified	v2: automatic check once billing data is in PadCareOne
Owner of site master data unknown `[A-18]`	Confirm with PadCare; default = Ops City Manager

9. Final design review

Before submission every part was cross-checked against the others. Beyond the walkthrough findings (Section 5.3), writing the test cases and a consistency audit found and fixed:

- **A 3rd billing complaint would have become Hygiene-Critical** through the recurrence escalator — meaningless for an invoice. Escalators now apply to operational complaints only; billing escalates instead.
- **The "3+ sites" escalator would have made #5 Hygiene-Critical** with no hygiene exposure — same fix.
- **"Recurrence" meant two things** (a link from the 2nd case, a flag from the 3rd), so the repeat rate and the warning counted differently. Now: *repeat* from the 2nd case, *Recurrence flag* from the 3rd.
- **"Customer not found → block"** would have lost issues from new or misspelled customers; replaced by an unmatched-customer placeholder that blocks only triage.
- Closure reasons, step numbers and routing text unified across all sections.

Check result: all five sample complaints run end to end; all eight challenges map to specific steps (two partly by design); every PadCare capability used is listed as an assumption.